

Predicting Employee Burnout and Resignation Risk Using Machine Learning on Workplace Behavioral Data

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Problem Statement

Organizations often discover employee dissatisfaction or burnout only after it escalates into resignation, by which point retention is no longer possible. Most HR teams lack accessible, data-driven tools to proactively identify at-risk employees using data they already collect, such as attendance, overtime, and satisfaction metrics. This project develops a predictive system that classifies employees into resignation-risk categories and estimates related outcomes such as tenure, enabling HR teams to intervene early.

Dataset Source and Justification

The IBM HR Analytics Employee Attrition & Performance dataset (Kaggle) was selected. It contains 1,470 employee records with 35 features directly aligned with the project proposal — including OverTime, JobSatisfaction, WorkLifeBalance, MonthlyIncome, and YearsAtCompany — with Attrition as the target variable representing resignation. Its combination of behavioral, demographic, and tenure features makes it well suited for both the classification (resignation prediction) and regression (tenure/satisfaction prediction) components of the proposal.

Dataset Overview

- Dimensions: 1,470 rows × 35 columns (31 after removing constant/identifier columns)
- Data types: 26 numeric, 9 categorical (text) attributes
- Target variable: Attrition (Yes/No) — 16.1% Yes, 83.9% No (class imbalance present)
- Missing values: 0 | Duplicate records: 0
- Outliers detected in MonthlyIncome, TotalWorkingYears, and YearsAtCompany, representing genuine senior/long-tenured employees rather than data errors

Feature Descriptions

Feature	Description
Age	Employee's age
Attrition	Target variable — whether the employee resigned or not (Yes/No)
BusinessTravel	Frequency of business travel required for the job
Department	Department the employee works in
DistanceFromHome	Distance between home and office (in km)
Education	Education level (1 = Below College, 5 = Doctorate)
EducationField	Field of study
EnvironmentSatisfaction	Satisfaction with work environment (1-4)
Gender	Employee's gender
JobInvolvement	Level of involvement in job (1-4)
JobLevel	Job level within the company
JobRole	Employee's job role
JobSatisfaction	Job satisfaction score (1-4)
MaritalStatus	Marital status
MonthlyIncome	Monthly salary
NumCompaniesWorked	Number of companies worked at previously
OverTime	Whether the employee works overtime (Yes/No)
PercentSalaryHike	Percentage increase in salary
PerformanceRating	Performance rating (1-4)
RelationshipSatisfaction	Satisfaction with workplace relationships (1-4)
StockOptionLevel	Level of stock options granted
TotalWorkingYears	Total years of professional experience
TrainingTimesLastYear	Number of training sessions attended last year
WorkLifeBalance	Work-life balance rating (1-4)
YearsAtCompany	Number of years at the current company
YearsInCurrentRole	Number of years in the current role
YearsSinceLastPromotion	Years since the last promotion
YearsWithCurrManager	Years spent with the current manager

Preprocessing Steps Performed

- Removed 4 non-informative columns: EmployeeCount, StandardHours, Over18 (constant values) and EmployeeNumber (identifier only)
- Verified dataset for missing values and duplicate records — none found, no imputation required
- Outlier inspection via boxplots; outliers retained as they represent valid senior-employee data relevant to attrition prediction
- Label Encoding applied to binary categorical features: Attrition, Gender, OverTime

- One-Hot Encoding applied to multi-class categorical features: BusinessTravel, Department, EducationField, JobRole, MaritalStatus (44 features after encoding)
- Feature Scaling: StandardScaler used to standardize all numeric features (mean = 0, SD = 1), justified by the large scale differences between features such as MonthlyIncome (1K-20K) and rating-scale features (1-4), which would otherwise bias distance- and gradient-based models
- Train-test split: 80% training (1,176 rows) / 20% testing (294 rows), stratified to preserve class ratio
- Class Imbalance: Training set was imbalanced (986 No / 190 Yes). SMOTE-Tomek was applied on the training set only to generate synthetic minority-class samples and remove overlapping boundary points, resulting in a balanced 986/986 distribution.

Libraries and Tools Used

Python, Pandas, NumPy, Scikit-learn (StandardScaler, train_test_split), Imbalanced-learn (SMOTETomek), Matplotlib, Google Colab, Git & GitHub.

Challenges Encountered

The dataset required no missing-value handling, simplifying preprocessing. The main challenge was the pronounced class imbalance (~5:1) in the target variable, addressed using SMOTE-Tomek applied strictly to the training split to avoid test-set contamination. Deciding how to treat outliers in tenure and income also required care, since they reflect genuine senior employees rather than data-entry errors.